

Thomas Gonzalez

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EDUCATION

Johns Hopkins University
MS in Robotics
BS in Mechanical Engineering
GPA: 3.65/4.0

Baltimore, MD
Expected May 2028
Expected May 2027

TECHNICAL SKILLS

- **CAD Tools:** SolidWorks (300+ hours), Autodesk Fusion 360 (50+ hours), Creo (50+ hours)
- **Programming & Software:** MATLAB (200+ hours), Java (100+ hours), Python (50+ hours), C++ (50+ hours), Ansys FEA (50+ hours)
- **Prototyping & Fabrication:** DFM, Root Cause Analysis, Soldering, Milling & Lathe Operation, 3D Printing
- **Language Skills:** English (fluent), Spanish (fluent), German (fluent)

INTERNSHIP EXPERIENCE

R&D MECHANICAL ENGINEERING INTERN

Thorlabs Imaging Systems

Sterling, VA
May 2026 – Present

- Led ideation, design, and launch various adapter plate SKUs for cage system platforms, expanding product offerings to better meet customer needs.
- Created first full assembly of a Bergamo multiphoton microscope for internal and external use.
- Manufactured custom aluminum jigs applying DFM principles and using CNC Mill to improve various manufacturing processes.
- Assembled, aligned and calibrated advanced microscopy systems for customers.

ENGINEERING INTERN

The Maryland Zoo in Baltimore

Baltimore, MD
April 2025 – May 2026

- Enhanced animal well-being through animal enrichment devices, integrating Arduino microcontrollers with sound, motion detection, and RF communication modules.
- Developed a C++ user interface to streamline usability for zookeepers and simplify system control.
- Achieved 95% reading detection accuracy through testing and validation of sensors, including LiDAR, PIR, and Doppler.
- Established partnerships with leading Zoos nationwide, resulting in plans to implement enrichment devices across 5+ states.

RESEARCH INTERN

AMIRo Lab - Laboratory for Computational Sensing and Robotics (LCSR)

Baltimore, MD
March 2025 – August 2025

- Developed calibration software to measure and predict needle deflection during robotic prostate cancer surgeries.
- Built a manual calibration rig in Creo to support data collection and system characterization for algorithm development.
- Programmed and operated a 4-DOF mechanical arm in ROS2 and Python to perform gel-phantom needle insertion experiments.
- Validated algorithm performance by comparing predicted needle geometry against ground-truth stereo-camera measurements.

LEADERSHIP EXPERIENCE

MECHANICAL SUBTEAM LEAD

Hopkinauts Mars Rover Team

Baltimore, MD
January 2024 - Present

- Led a team of 20+ students in designing and building the frame, robotic arm, and science box for the University Rover Challenge.
- Designed a 6-DOF robotic arm using SolidWorks equipped with custom cycloidal gearboxes and a precision end-effector.
- Manufactured precision welded steel frame maximizing rigidity and dust resistance while minimizing weight.
- Achieved 80% life detection accuracy with custom soil delivery mechanism, chemical analyses, and spectrometer.
- Raised \$25k through sponsorships and donations, presented project work at JHU Design Day 2026 and JHU Physics Day 2026.

RECRUITMENT CHAIR

Society of Hispanic Professional Engineers

Baltimore, MD
February 2024 – May 2026

- Created the Familia Mentorship program, pairing incoming Hispanic students with upperclassmen to facilitate college transition.
- Maintained community and promoted cultural exchange through social and professional events.
- Helped grow chapter membership by over 50% by forming connections and promoting SHPE resources at events, in classes, and at fairs.